

Quiz2

CSC354 · Machine Learning

TOTAL MARKS
4

DURATION
8 min

DIFFICULTY
HARD

PASSING THRESHOLD
75%

STUDENT NAME

STUDENT ID

SECTION

DATE

INSTRUCTIONS

- All questions are compulsory.
- Total time: 8 minutes.
- Total marks: 4.

Questions

CLO-1 Illustrate various concept learning algorithms with suitable examples.

1. Which of the following algorithms is best suited for learning non-linear decision boundaries in high dimensional spaces, often used as a preprocessing step before applying Support Vector Machines (SVM)?

- a) Linear Regression
- b) Decision Trees with Pruning
- c) Kernel Principal Component Analysis (KPCA)
- d) Naive Bayes Classifier

2. Which of the following algorithms is most suitable for learning a decision tree classifier with minimal overfitting?

- a) Option A - Decision Tree Algorithm without pruning.
- b) Option B - Random Forest Classifier using bagging and feature randomness to improve generalization.
- c) Option C
- d) Option D

3. Which of the following algorithms is most suitable for a scenario where you have to learn from incomplete labeled data?

- a) Option A
- b) Option B
- c) Option C
- d) Option D

4. Which of the following algorithms is most suitable for a classification problem with imbalanced classes?

- a) Option A - Decision Trees without any pre-processing.
- b) Option B - Synthetic Minority Over-sampling Technique (SMOTE).
- c) Option D
- d) Option D